

VITA: ZERO-SHOT VALUE FUNCTIONS VIA TEST-TIME ADAPTATION OF VISION-LANGUAGE MODELS

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ABSTRACT

Vision-Language Models (VLMs) show promise as zero-shot goal-conditioned value functions, but their frozen pre-trained representations limit generalization and temporal reasoning. We introduce VITA, a zero-shot value function learning method that enhances both capabilities via test-time adaptation. At inference, a lightweight adaptation module is updated via a gradient step on a meta-learned self-supervised loss, such that each test-time update improves value estimation. By updating sequentially over a trajectory, VITA encodes history into its parameters, addressing the temporal reasoning limitations. To mitigate shortcut learning, we propose a dissimilarity-based sampling strategy that selects semantically diverse segments of the trajectory during training. Trained on a single real-world environment and evaluated across ten out-of-distribution robotic manipulation datasets spanning task, environment, and embodiment shifts, VITA attains the strongest value-order correlation on 6 of the 10 datasets and roughly triples the score of the state-of-the-art autoregressive-VLM method (e.g., 0.782 vs. 0.269 in-distribution), while never falling below it on any shift. Its advantage is most pronounced on the hardest long-horizon sweeping task, where every competing method degrades sharply. Used for reward shaping in offline reinforcement learning, VITA’s zero-shot value estimates yield multi-task policies on the Meta-World MT10 benchmark that reach an interquartile-mean return of 0.815, surpassing every CLIP-based baseline as well as the simulation’s hand-designed fuzzy-logic dense reward (0.779), with non-overlapping confidence.

1 INTRODUCTION

Vision-Language Models (VLMs) have demonstrated strong generalization across diverse tasks and domains by learning from large-scale, unstructured web data without human supervision (Radford et al., 2021; Alayrac et al., 2022). In contrast, despite significant advances in learning generalist policies for robotics (Brohan et al., 2023; Kim et al., 2024) and 3D virtual environments (Reed et al., 2022; SIMA Team et al., 2024), state-of-the-art methods have yet to achieve comparable success, largely because they still depend on expert demonstrations (Ho & Ermon, 2016). World models (Ha & Schmidhuber, 2018; Hafner et al., 2021; Matsuo et al., 2022) have emerged as a promising solution, enabling agents to learn in simulated environments, with applications spanning robotics (Yang et al., 2023; Agarwal et al., 2025), autonomous driving (Hu et al., 2024), and video games (Bruce et al., 2024). Yet even as world models generate increasingly realistic visual trajectories in diverse, open-ended environments (Hughes et al., 2024; Silver & Sutton, 2025), one critical challenge remains unresolved: *how can agents effectively learn from videos at scale without relying on human supervision?*

A line of research explores policy learning directly from expert visual trajectories by inferring actions from visual transitions (Edwards et al., 2019; Baker et al., 2022; Schmidt & Jiang, 2024; Zhang et al., 2025a). These latent actions can be mapped to executable controls in deployment, although this remains a challenging task in practice. In parallel, other work focuses on learning a universal *goal-conditioned value function*, using it as a zero-shot reward shaping and supervision signal for reinforcement learning (RL) and imitation learning, respectively (Chen et al., 2021; Ma et al., 2022). In this framework, goal-conditioned value estimation is formulated as predicting how far an agent has progressed toward completing a task, given only visual observations and a natural language task description (Lee et al., 2021a; Ma et al., 2023b; 2024; Dashora et al., 2025). The quality of such an

estimator is naturally quantified by how well its predicted progress tracks the true temporal order of a trajectory, a measurable criterion we return to throughout our evaluation.

Prior work has employed pre-trained contrastive VLMs (Radford et al., 2021) for zero-shot reward shaping, leveraging the similarity between task descriptions and visual observations in a shared multimodal representation space to ground progress in semantic context (Ma et al., 2023b; Baumli et al., 2023; Rocamonde et al., 2024). However, these methods fail to capture the temporal context needed to disambiguate visually similar states that occur at different stages of a task (e.g., folding vs. unfolding a shirt); as our experiments quantify, such contrastive-only estimators barely correlate with true task progress, and in some out-of-distribution scenes their correlation even turns negative. In contrast, autoregressive VLMs (Alayrac et al., 2022) incorporate temporal context by conditioning on the entire visual trajectory within the prompt; however, they inherit a bias toward monotonically increasing predictions from the chronologically ordered datasets used in pre-training (Ma et al., 2024), which we find caps their correlation well below temporally aware learners across every shift we test. Both VLM architectures rely on pre-trained representations (Ma et al., 2023b; 2024; Baumli et al., 2023; Rocamonde et al., 2024; Du et al., 2023) for zero-shot prediction, which limits both their generalization and their temporal reasoning (Pătrăucean et al., 2023). To enable generalizable value function learning, recent work has explored large-scale pre-training (Alakuijala et al., 2024; Ma et al., 2023b), domain-specific fine-tuning (Zhang et al., 2025b), and demo-based reward specification (Sontakke et al., 2023).

In this work, we introduce VITA, a zero-shot value function learning method that enhances both generalization and temporal reasoning through test-time adaptation, and that consistently outperforms the state-of-the-art zero-shot approach (Ma et al., 2024) on real-world robotic manipulation tasks. Unlike prior methods that rely on large-scale pre-training or expert demonstrations for generalization, VITA adapts online to both the semantic and temporal context of trajectories. At inference, a lightweight adaptation module is updated in negligible time with a gradient step on a meta-learned self-supervised loss (Sun et al., 2024), such that each test-time update improves value function estimation (Finn et al., 2017). By updating sequentially over a test trajectory, the value function estimator encodes trajectory history into its parameters (Sun et al., 2024), thereby addressing the temporal reasoning limitations of prior work. To mitigate shortcut learning (Geirhos et al., 2020), we introduce a dissimilarity-based sampling strategy that encourages reliance on semantic cues; our ablation shows it lifts expert-vs.-suboptimal discrimination to a perfect score, well above competing sampling strategies. Across three complementary evaluations, VITA is the strongest or statistically tied-strongest method on each: generalization under distribution shifts (best value-order correlation on 6 of 10 datasets), differentiation between expert and non-expert trajectories (perfect discrimination), and reward shaping for offline RL (Levine et al., 2020) (highest interquartile-mean return).

We summarize our key contributions as follows:

- We propose VITA, a test-time adaptation method that enhances the generalization and temporal reasoning of contrastive VLMs for zero-shot value function estimation, without requiring task-specific demonstrations or large-scale pre-training.
- Trained on a single environment and evaluated on ten out-of-distribution datasets spanning task, environment, and embodiment shifts, VITA achieves the best value-order correlation on 6 of the 10, roughly triples the correlation of the state-of-the-art zero-shot method (Ma et al., 2024), and never scores below it on any shift.
- Using VITA’s zero-shot value estimates for reward shaping yields offline RL policies on the Meta-World multi-task benchmark (MT10) with an interquartile-mean return of 0.815, exceeding every CLIP-based baseline and even the simulation’s hand-designed fuzzy-logic dense reward (0.779) (Yu et al., 2020), with non-overlapping 95% confidence intervals.

2 PRELIMINARIES

2.1 GOAL-CONDITIONED VALUE FUNCTIONS

In video trajectories, task progress estimation can be viewed as goal-conditioned value function estimation, where the reward reflects task completion. We therefore formulate learning a goal-conditioned value function as predicting the degree of task completion from vision–language representations.

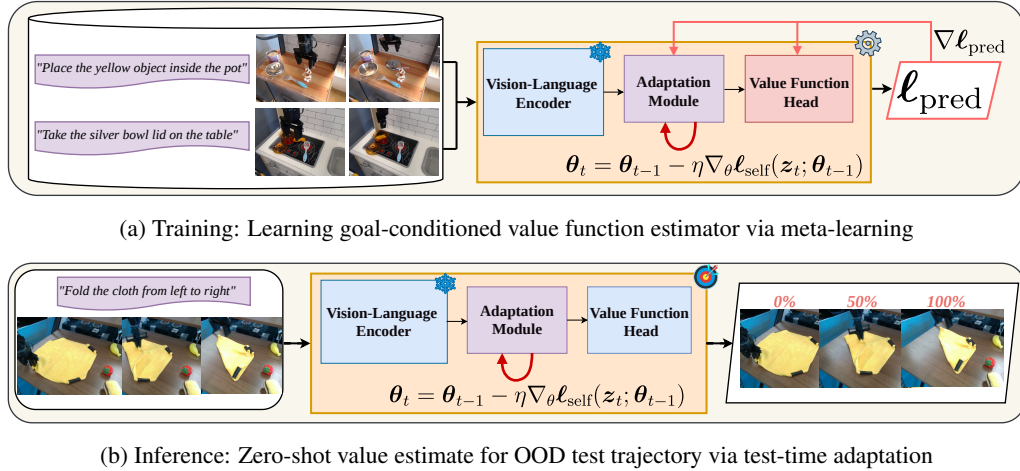


Figure 1: **Overview of VITA.** VITA learns a goal-conditioned value function via meta-learning and achieves zero-shot generalization to out-of-distribution trajectories via test-time adaptation.

Formally, the value function is defined as $V : \mathcal{O} \times \mathcal{G} \rightarrow [0, 1]$, mapping an observation $o_t \in \mathcal{O}$ and a goal specification $g \in \mathcal{G}$ to a scalar that quantifies predicted progress toward goal completion, with $V(o_t; g) = 0$ marking the start of the task and $V(o_t; g) = 1$ marking completion. Task progress is commonly aligned with temporal position in expert demonstrations, based on the assumption that such trajectories exhibit monotonically increasing progress toward goal completion (Lee et al., 2021a; Ma et al., 2024; Dashora et al., 2025). Given an expert trajectory $\tau = (o_1, \dots, o_T) \sim \pi_E$, temporal progress is defined using normalized timestep indices: $V^{\pi_E}(o_t; g) = \frac{t}{T}$ where T is the trajectory length. This normalized-timestep target provides the supervision signal for learning a goal-conditioned value function V , and, as we detail in the experiments, the degree to which a learned estimator recovers this ordering is exactly what our value-order-correlation metric measures.

2.2 TEST-TIME TRAINING FOR SEQUENCE MODELING

Test-time training (TTT) (Sun et al., 2020) is a test-time adaptation method that treats inference as a self-supervised learning task (Goodfellow et al., 2016), updating model parameters on each test instance without access to labels. While originally proposed for static image classification (Sun et al., 2020), recent work extends TTT to sequence modeling (Sun et al., 2024; Wang et al., 2023), where a parametric adaptation module f_{adapt} is updated at each timestep using a gradient step on a self-supervised loss. In contrast to standard sequence architectures that encode temporal history in activation states, such as hidden states (Hochreiter & Schmidhuber, 1997; Cho et al., 2014) or cached key-value pairs (Vaswani et al., 2017), TTT encodes history into the parameters of f_{adapt} via sequential updates at inference. This distinction is not merely conceptual: our controlled comparison against a hidden-state recurrent variant (CLIP-GRU) and against batched-update TTT variants shows that where history is stored has a large, measurable effect on value estimation. The parameters of f_{adapt} thereby serve as an implicit memory, encoding temporal history while preserving temporal order. Sun et al. (2024) further propose to meta-learn the self-supervised task, following the gradient-based meta-learning paradigm (Finn et al., 2017). In this setting (Sun et al., 2024), the self-supervised loss is parameterized by learnable linear projections and trained to minimize downstream prediction loss after a test-time update, rather than directly minimizing it. In this work, we apply TTT to goal-conditioned value function learning, where temporal context is captured through sequential updates, and the self-supervised task is meta-learned to improve value estimation rather than being predefined a priori.

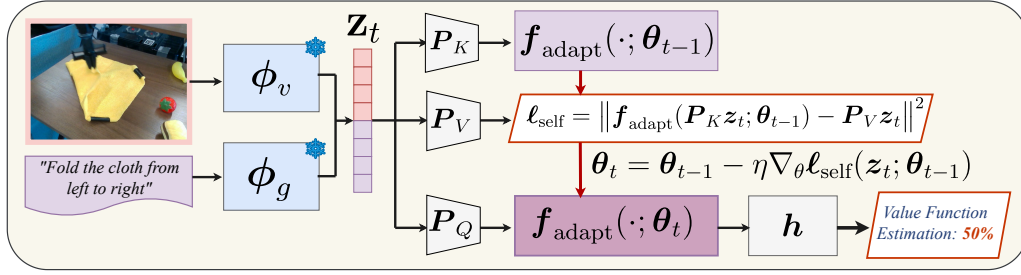


Figure 2: **Test-time Adaptation.** In inference, at each timestep t , an adaptation module f_{adapt} is updated via a gradient step on a meta-learned self-supervised loss ℓ_{self} , encoding temporal history.

3 VITA: ZERO-SHOT VALUE FUNCTIONS VIA TEST-TIME ADAPTATION

3.1 MODEL ARCHITECTURE

Our goal-conditioned value function estimator comprises three modules: (1) a multimodal encoder that extracts joint visual-language representations from visual trajectories and their task descriptions; (2) an adaptation module updated at test-time using a self-supervised loss that is meta-learned to improve value estimation; and (3) a regression head that predicts value estimates.

3.1.1 MULTIMODAL INPUT REPRESENTATION

We use a frozen contrastive vision-language encoder, CLIP (Radford et al., 2021), to extract representations from visual observations and goal descriptions. Given a visual trajectory $\tau = (o_1, \dots, o_T)$ and a language task description g , we concatenate their representations at each timestep t to form a sequence of joint multimodal representations $(z_1, z_2, \dots, z_T)$, where $z_t = [\phi_v(o_t); \phi_g(g)] \in \mathbb{R}^{2d}$, and ϕ_v and ϕ_g denote the visual and language encoders, respectively. CLIP is pre-trained with a contrastive objective to align paired image-text inputs in a shared representation space (Radford et al., 2021), so that visual observations semantically closer to goal completion sit closer to the goal-description representation. We adopt this concatenation-based joint representation over an element-wise product because it yielded consistently higher validation performance (Appendix D).

3.1.2 TEST-TIME ADAPTATION

To adapt multimodal representations to both semantic and temporal context, we employ an adaptation module f_{adapt} following the test-time training paradigm described in 2.2. At inference, the parameters of f_{adapt} are updated online via a gradient step on a meta-learned self-supervised loss ℓ_{self} . This loss is formulated as a reconstruction objective using learnable linear projections P_K and P_V , which are meta-learned during training to improve value estimation after test-time adaptation. In particular, given $P_K \in \mathbb{R}^{d' \times 2d}$ that generates a perturbed input view and $P_V \in \mathbb{R}^{d' \times 2d}$ for the target, the self-supervised loss is defined as:

$$\ell_{\text{self}}(z_t; \theta_{t-1}, P_K, P_V) = \|f_{\text{adapt}}(P_K z_t; \theta_{t-1}) - P_V z_t\|^2. \quad (1)$$

where θ_{t-1} denotes the parameters of f_{adapt} before test-time adaptation. At each timestep t , the parameters are updated by

$$\theta_t = \theta_{t-1} - \eta \nabla_{\theta} \ell_{\text{self}}(z_t; \theta_{t-1}), \quad (2)$$

with learning rate η . Through sequential updates, f_{adapt} implicitly retains information from past visual observations, thereby capturing temporal context. In Section 4.6.2, we quantify this design choice directly: incremental per-timestep updates raise validation VOC by roughly 0.5–0.6 over trajectory-level, memoryless, and explicit-window updates on every dataset, confirming that when and how often the estimator adapts matters as much as the adaptation itself.

3.1.3 ZERO-SHOT VALUE FUNCTION ESTIMATOR

After test-time adaptation, a meta-learned linear projection $P_Q \in \mathbb{R}^{d' \times 2d}$ maps the input z_t into an adaptation space $\mathbb{R}^{d'}$. Then, this representation is passed through the adaptation module f_{adapt} , followed by a regression head h that outputs predicted values of the goal-conditioned value function:

$$V(z_t; g) = h(f_{\text{adapt}}(P_Q z_t; \theta_t)) \quad (3)$$

The regression head h is a two-layer multilayer perceptron (MLP), following value-function estimator architectures used in deep reinforcement learning (Espeholt et al., 2018). The network is trained using normalized progress labels y_t from expert demonstrations as described in Section 2.1, with a supervised prediction loss ℓ_{pred} defined as the mean squared error between predicted values $V(z_t; g)$ and targets y_t . At inference, the regression head h remains frozen while only the lightweight adaptation module f_{adapt} is updated online using the self-supervised objective. Because only this small module changes at test time, the per-timestep overhead is negligible, yet it is enough to let the estimator generalize zero-shot to unseen tasks, environments, and embodiments without any task-specific demonstrations.

3.2 TRAINING PROCEDURE

We train the value function estimator with gradient-based meta-learning, optimizing a self-supervised loss ℓ_{self} such that test-time adaptation improves the supervised prediction loss ℓ_{pred} (Sun et al., 2024). Training is performed on diverse sub-trajectories selected via dissimilarity-based sampling, which mitigates shortcut learning and encourages reliance on semantic and temporal cues; in our ablation (Section 4.6.1) this single design choice is what separates perfect expert-vs.-suboptimal discrimination from the partial discrimination achieved by full-trajectory or random sampling.

3.2.1 META-LEARNING

During training, the adaptation module f_{adapt} is updated online at each timestep using the self-supervised loss ℓ_{self} . Following the gradient-based meta-learning paradigm (Finn et al., 2017), we differentiate through these adaptation updates with respect to the supervised prediction loss ℓ_{pred} . This optimizes the initialization θ_0 of f_{adapt} , the linear projections P_K, P_V, P_Q , and the regression head h . Formally, the total training loss combines the supervised prediction loss ℓ_{pred} with the self-supervised loss ℓ_{self} , weighted by a scalar λ . The learned parameters θ_0 serve as the initialization for adaptation at inference time. This approach enables the value function estimator to learn to adapt its internal representations at test time to improve value function estimation.

3.2.2 DISSIMILARITY-BASED SAMPLING

Expert visual trajectories often contain consecutive frames that are highly redundant, as noted in prior work on video action recognition (Wang et al., 2018). Such redundancy can encourage the value function estimator to exploit shortcut cues (Geirhos et al., 2020; Lee et al., 2021b) by overfitting to frequently occurring late-stage visual patterns. To mitigate this, we propose a dissimilarity-based sampling strategy that constructs mini-batches from the most visually diverse sub-trajectories within each trajectory. This increases intra-batch variance and acts as a form of importance sampling that favors semantic diversity, and it does so at negligible cost relative to model training (Appendix G).

Formally, given a sequence of multimodal representations $\{z_1, \dots, z_T\}$, we extract fixed-length sub-trajectories using a sliding window of size w_{tr} and stride s , yielding a candidate set $\mathcal{W}$. A diverse subset of size k maximizes pairwise dissimilarity:

$$\mathcal{W}' = \arg \max_{\mathcal{U} \subset \mathcal{W}, |\mathcal{U}|=k} \sum_{\{w^i, w^j\} \in \binom{\mathcal{U}}{2}} \|w^i - w^j\|_2^2, \quad (4)$$

where each w^i is a sub-trajectory of length w_{tr} . Solving Eq. 4 is intractable, requiring enumeration of all $\binom{|\mathcal{W}|}{k}$ subsets. To approximate this objective efficiently, we adopt a scoring-based heuristic, assigning each window $w \in \mathcal{W}$ a score equal to its total dissimilarity with all others:

$$s(w) = \sum_{v \in \mathcal{W}} \|w - v\|_2^2, \quad \mathcal{W}' = \arg \max_{\mathcal{U} \subset \mathcal{W}, |\mathcal{U}|=k} \sum_{w \in \mathcal{U}} s(w). \quad (5)$$

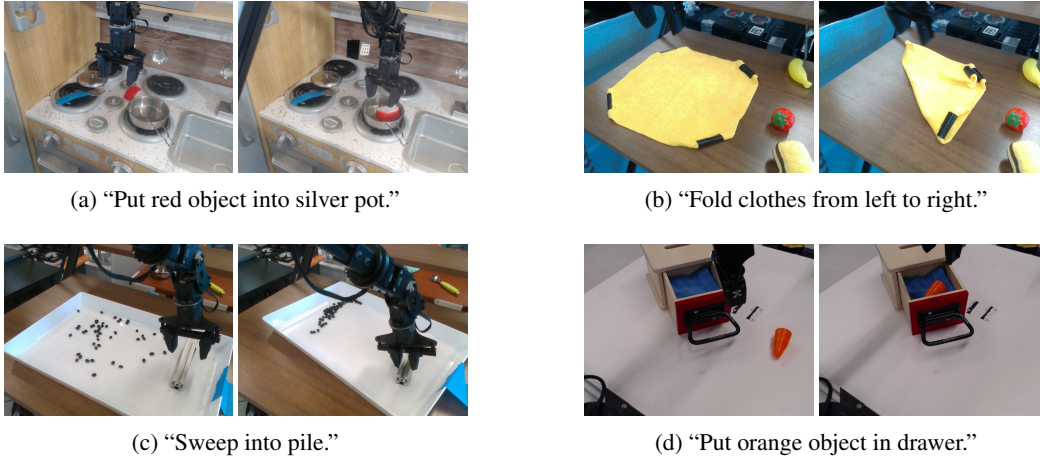


Figure 3: Representative visual trajectories paired with task descriptions, spanning the increasingly severe distribution shifts on which we evaluate: (a) in-distribution pick-and-place; (b, c) environment shift to unseen folding and sweeping scenes; (d) combined embodiment and environment shift. Despite training only on the setting in (a), VITA maintains strong value-order correlation across all of these conditions (Table 1).

Thus, by computing a diversity score s for each window and selecting the k highest-scoring windows, we obtain a heuristically diverse subset with polynomial-time complexity. Our complexity analysis in Appendix G shows the resulting overhead is negligible, on the order of a few hundred MFLOPs, far below the cost of a single forward-backward pass. When we compare this strategy head-to-head against full-trajectory and random sampling in the ablation study (Section 4.6.1), it improves expert-vs.-non-expert discrimination from as low as 0.20 up to a perfect 1.0.

4 EXPERIMENTS

We organize our evaluation around three core capabilities of goal-conditioned value functions, and report a quantitative comparison against six baselines on each: (i) generalization in value estimation for real-world robotic manipulation under distribution shifts in task, environment, and embodiment, measured by value-order correlation across 10 datasets; (ii) differentiation between expert and non-expert visual trajectories in real-world robotic manipulation tasks, measured by binary discrimination accuracy across 5 scripted datasets; (iii) zero-shot reward shaping for offline RL in Meta-World MT10, a simulated multi-task benchmark, measured by interquartile-mean return with bootstrap confidence intervals. Across all three, VITA is the top-performing or statistically tied-top method, and the following subsections quantify the margins by which it does so.

4.1 TRAINING SETUP

We train VITA on expert visual trajectories paired with natural language task descriptions from the BridgeData V2 dataset (Walke et al., 2023), which spans a wide range of manipulation tasks, environments, and robot embodiments. Notably, our training set is deliberately narrow: a curated subset of 2,986 expert demonstrations that covers only pick-and-place manipulation, with no folding, sweeping, or stacking tasks, collected from a single robot embodiment (WidowX 250) across just 4 configurations of the ToyKitchen environment. This narrow training distribution makes the breadth of the out-of-distribution evaluation that follows all the more demanding. An additional 287 expert demonstrations are held out as the in-distribution test set, referred to as `tk_pnp`. Appendix A includes examples from the training set. We use OpenCLIP ViT-B/32 as the frozen backbone for encoding video frames and task descriptions. The training objective combines ℓ_{pred} with ℓ_{self} , weighted by $\lambda_{\text{self}} = 0.5$. During training, we use dissimilarity-based sampling with window size $w_{\text{tr}} = 8$, number of sub-trajectories $k = 8$, and stride $s = 1$. At test time, we use one gradient step ($t_{\text{ep}} = 1$) using λ_{self} with learning rate $\eta = 0.1$. Further details and hyperparameter sweeps are reported in Appendix D.

4.2 BASELINES

We benchmark VITA against six value functions that together span the two dominant VLM-based paradigms, contrastive and autoregressive, so that each comparison isolates a specific design axis. Two contrastive, temporally-agnostic baselines establish a floor: VLM-CL computes zero-shot value estimates from cosine similarity between CLIP representations of each frame and task description (Baumli et al., 2023), and VLM-RM regularizes CLIP embeddings by projecting features along the direction from a generic reference prompt to the task prompt (Rocamonde et al., 2024). CLIP-FT adds supervised training by fitting a regression head on frozen CLIP multimodal representations, isolating the effect of supervision without temporal modeling. CLIP-GRU instead adds temporal modeling by training a Gated Recurrent Unit (GRU) (Cho et al., 2014) on the same frozen CLIP representations; it is the most direct point of comparison for VITA, differing only in how temporal history is stored (recurrent hidden states vs. adapter parameters). Finally, GVL (Ma et al., 2024), the state-of-the-art zero-shot value function built on autoregressive VLMs, appears in two configurations: GVL-0S, the zero-shot setting where the VLM is prompted with only the test trajectory and task description, and GVL-1S, a one-shot in-context setting that additionally provides a full shuffled training-distribution trajectory with its progress labels. Following GVL, we use Gemini 1.5 Pro (gemini-1.5 Pro-latest) (Gemini Team et al., 2024) with its proposed prompt template. For completeness we also tested two alternative VLMs under the same protocol: Qwen-VL 2.5 (Yang et al., 2024), which failed to overcome temporal bias, and GPT-4o (Hurst et al., 2024), which refused to produce value estimates. Additional implementation details are provided in Appendix D.

4.3 EVALUATING GENERALIZATION UNDER DISTRIBUTION SHIFTS

Table 1: Value-Order Correlation (VOC; higher is better) for value function estimation under distribution shifts, across all seven methods and ten datasets. The best score per row is in **bold** and the second-best is underlined. VITA attains the best VOC on 6 of the 10 datasets and is second-best on the remaining 4; on every row it exceeds the state-of-the-art autoregressive baselines GVL-0S and GVL-1S by a wide margin (e.g., 0.782 vs. 0.269 in-distribution). Shift regimes: **ID** = In-Distribution, **ES** = Environment Shift, **EM** = Embodiment Shift, **ES & EM** = Both Shifts.

Shift	Dataset	VLM-CL	VLM-RM	CLIP-FT	GVL-0S	GVL-1S	CLIP-GRU	VITA
ID	tk_pnp	0.038	0.029	0.251	0.269	0.252	<u>0.773</u>	0.782
ES	lm_pnp	0.017	0.033	0.149	0.305	0.272	<u>0.676</u>	0.725
	td_fold	0.031	0.072	0.152	0.326	0.318	<u>0.674</u>	0.709
	ft_fold	0.108	0.099	0.162	0.331	0.387	0.693	<u>0.658</u>
	rd_fold	0.095	0.055	0.126	0.372	0.406	0.726	<u>0.606</u>
	ms_sweep	-0.129	-0.226	0.148	0.158	0.150	<u>0.434</u>	0.490
EM	dt_tk_pnp	0.042	-0.041	0.149	0.258	0.211	0.856	<u>0.820</u>
	dt_tk_stack	0.035	0.046	0.099	0.254	0.277	<u>0.667</u>	0.708
ES & EM	dt_ft_stack	0.026	0.028	0.049	0.212	0.249	<u>0.674</u>	0.698
	dt_rd_pnp	0.023	0.041	0.211	0.329	0.316	0.747	<u>0.695</u>

We evaluate how well progress estimators generalize across novel environments, tasks, and robot embodiments using subsets of BridgeData V2 curated to introduce variation along each axis, all held out from the pick-and-place-only training set. Environment shifts involve changes in scene layout or background: `lm_pnp` is a pick-and-place task in front of a laundry machine, `td_fold`, `ft_fold`, and `rd_fold` feature cloth folding on different surfaces, and `ms_sweep` introduces a long-horizon sweeping task in a confined tray. Embodiment shifts use the DeepThought robot, which differs from the training embodiment (WidowX 250) in both morphology and camera perspective: `dt_tk_pnp` (pick-and-place) and `dt_tk_stack` (stacking) keep the in-distribution environment but change the embodiment, while `dt_ft_stack` (stacking) and `dt_rd_pnp` (drawer pick-and-place) combine both shifts. Each dataset comprises 200 expert trajectories with task descriptions, except `ms_sweep`, which contains 100. Appendix B provides a full description. We quantify performance with the *Value Order Correlation* (VOC) metric (Ma et al., 2024), which measures how well predicted progress tracks the chronological order of frames: letting $p_1, \dots, p_T$ denote the predicted progress values for a T -frame trajectory and $r_t = t$ the temporal index, VOC is the Spearman rank correlation ρ_s between

the two, so that a score of 1 indicates perfectly ordered predictions and 0 indicates no better than chance.

Table 1 reveals a clear and consistent ordering of the seven methods. The two temporally-aware learners, VITA and CLIP-GRU, dominate both autoregressive baselines GVL-0S and GVL-1S on every one of the 10 datasets, typically by more than 0.4 VOC (e.g., 0.782 vs. 0.269 in-distribution, and 0.725 vs. 0.305 on `lm_pnp`). This gap underscores both the limitations of pre-trained autoregressive VLMs and the value of preserving temporal order for progress estimation. Between the two temporally-aware methods, VITA is the stronger: it wins on 6 of the 10 datasets and is a close second on the other 4, with the largest per-dataset margins in its favor on the hardest shifts. Both VITA and CLIP-GRU in turn leave the temporally-agnostic CLIP baselines (CLIP-FT, VLM-CL, VLM-RM) far behind, isolating temporal history as the decisive factor: CLIP-FT matches GVL in-distribution (0.251 vs. 0.269) but collapses under every shift, while VLM-CL and VLM-RM hover near zero and even go negative on `ms_sweep` (-0.129 and -0.226). For GVL, adding a one-shot in-context example is inconsistent, sometimes helping, sometimes neutral, and sometimes hurting. GVL performs relatively well on folding tasks (`td_fold`, `ft_fold`, `rd_fold`) but drops sharply on stacking (`dt_tk_stack`, `dt_ft_stack`) and pick-and-place (`dt_tk_pnp`, `lm_pnp`), suggesting a bias toward folding-like manipulations. VITA, by contrast, stays uniformly high (VOC between 0.490 and 0.820) across every task type and shift, evidence of markedly more uniform generalization than in-context learning.

The long-horizon `ms_sweep` task is the hardest for every method except CLIP-FT, yet VITA still posts the highest VOC (0.490), edging out CLIP-GRU (0.434) while the CLIP-similarity baselines turn negative. That VITA leads precisely where the task is longest is direct evidence that encoding temporal history in adapter parameters via test-time adaptation scales to long horizons better than recurrent hidden states. Embodiment shift tells a similar story. On `dt_tk_pnp`, which changes only the robot while keeping the training environment and tasks, both VITA (0.820) and CLIP-GRU (0.856) actually surpass their own in-distribution scores, showing that learned temporal context transfers cleanly across embodiments. GVL-1S, by comparison, manages only 0.211 on `dt_tk_pnp` and 0.277 on `dt_tk_stack`, a shortfall of roughly 0.5–0.6 VOC that indicates few-shot in-context learning does not generalize under embodiment shift in our setup.

4.4 DIFFERENTIATING EXPERT FROM NON-EXPERT TRAJECTORIES

Beyond generalization, we probe a second capability, robustness, by testing whether VITA assigns lower progress scores to suboptimal behavior than to expert behavior. We compare model predictions on expert and scripted (non-expert) visual trajectories drawn from the same in-distribution setting of BridgeData V2. The scripted trajectories are generated in the ToyKitchen environment using a heavily randomized controller (Walke et al., 2023), under the same embodiment and environmental configuration as the expert demonstrations. They cover pick-and-place tasks over five object categories that may overlap with the training set: general objects (`sc_pnp_obj`), rigid items (`sc_pnp_robj`), soft toys (`sc_pnp_stoy`), utensils (`sc_pnp_uten`), and cloth (`sc_pnp_cloth`). Because these trajectories match the training distribution in task, embodiment, and environment yet deviate from the smooth, monotonic progress of expert demonstrations, they form a controlled test of whether an estimator tracks true progress rather than surface appearance. Appendix C includes examples. Since the randomized controller does not produce quantifiable degrees of suboptimality, we score discrimination in binary form: an effective estimator should simply rank expert trajectories above scripted ones. We formalize this with *BinVOC*, defined as $1 [\text{VOC}_{\text{exp}} > \text{VOC}_{\text{subopt}}]$, which equals 1 when the expert trajectory attains higher VOC than the suboptimal one and 0 otherwise, and report its average over the five scripted datasets.

Discriminative performance is reported as *BINVOC* averaged over all five scripted datasets (Table 2a), with expert scores (VOC_{exp}) measured on the in-distribution test set `tk_pnp`. Three methods, VITA, GVL-0S, and GVL-1S, achieve a perfect *BinVOC* of 1.00, correctly ranking expert above scripted on all five datasets. Crucially, VITA reaches this ceiling while CLIP-GRU, its closest architectural counterpart, stalls at 0.80, missing one dataset; this gap suggests that recurrent hidden states are more prone to overfitting temporal shortcuts than the implicit memory formed by sequential test-time updates. The temporally-agnostic CLIP baselines fare worst, with VLM-CL at 0.40 and VLM-RM at 0.00, consistent with their lack of temporal modeling. CLIP-FT is more reliable at 0.80 but still fails one task, pointing to sensitivity under object distribution shift.

Method	BinVOC
VLM-CL	0.40
VLM-RM	0.00
CLIP-FT	0.80
GVL-0S	1.00
GVL-1S	1.00
CLIP-GRU	0.80
VITA	1.00

(a) Expert vs. Non-Expert (BinVOC).

Method	IQM	95% CI
VLM-CL	0.760	[0.722, 0.791]
VLM-RM	0.746	[0.718, 0.771]
CLIP-FT	0.785	[0.759, 0.809]
CLIP-GRU	0.777	[0.734, 0.814]
META-WL	0.779	[0.750, 0.804]
VITA	0.815	[0.785, 0.838]

(b) Offline RL on MT10 (IQM).

Table 2: (a) Expert-vs.-non-expert discrimination, average BinVOC over 5 in-distribution scripted datasets (higher is better). VITA ties GVL-0S and GVL-1S at a perfect 1.00, while its closest architectural counterpart CLIP-GRU reaches 0.80. (b) Offline RL on Meta-World MT10, interquartile mean (IQM) of return with 95% stratified bootstrap CIs (higher is better). VITA leads at 0.815, surpassing all CLIP-based baselines and the simulator’s fuzzy-logic dense reward META-WL (0.779) with non-overlapping intervals. Best result per column in **bold**.

4.5 ZERO-SHOT REWARD SHAPING FOR OFFLINE RL

Our third evaluation asks whether VITA’s zero-shot value estimates are useful as shaped rewards for offline RL. We use the Meta-World MT10 benchmark (Yu et al., 2020), ten diverse robotic manipulation tasks including pick-and-place, door opening, and pushing, and stress the transfer gap by keeping VITA trained only on real-world robotic data while applying it zero-shot in this simulated setting. For each task we generate 20 expert visual demonstrations using Meta-World’s expert policies, use the goal-conditioned value estimator to assign a dense reward to each visual observation, and train policies with Implicit Q-Learning (IQL) (Kostrikov et al., 2021). Data generation and training are repeated over 10 random seeds; each resulting multi-task policy is evaluated on 20 rollouts per task, and we average return per (task, seed) pair. We then report the interquartile mean (IQM) over all pairs with 95% stratified bootstrap confidence intervals, following Agarwal et al. (2021). Full details appear in Appendix E.

The results (Table 2b) place VITA first with an IQM return of 0.815 (95% CI [0.785, 0.838]), ahead of every CLIP-based baseline: CLIP-FT (0.785), CLIP-GRU (0.777), VLM-CL (0.760), and VLM-RM (0.746). Most notably, VITA also surpasses META-WL (0.779), the simulator’s own hand-designed fuzzy-logic dense reward, by 0.036 IQM with confidence intervals that do not overlap, evidence that a value estimator trained purely on real-world data can supply reward signal that is competitive with, and here better than, an environment-specific engineered reward. A direct comparison with GVL is infeasible at this scale: it relies on the proprietary Gemini-1.5, whose inference cost is prohibitive in an RL loop, and the open-source alternatives we tested did not yield reliable progress estimates.

4.6 ABLATION STUDIES

4.6.1 EFFECT OF DISSIMILARITY-BASED SAMPLING ON DISCRIMINATIVE PERFORMANCE

We isolate the contribution of dissimilarity-based sampling by holding the model fixed and varying only the training-time sub-trajectory sampling strategy, then measuring the resulting expert-vs.-non-expert discrimination. Full-trajectory sampling (Sun et al., 2024), which computes the adaptation loss over all overlapping sub-trajectories with stride $s = 1$, overfits to global temporal shortcuts and discriminates worst. Random sampling improves diversity by drawing sub-trajectories uniformly but ignores semantic content. Dissimilarity-based sampling, which selects the sub-trajectories that maximize pairwise dissimilarity, explicitly promotes semantic diversity and is the only strategy to reach perfect discrimination, ranking above both alternatives. The full breakdown in Appendix F.1 shows the ordering concretely: average Disc@VOC rises from 0.20 (full-trajectory) to 0.80 (random) to 1.0 (dissimilarity-based).

4.6.2 EFFECT OF TEMPORAL MEMORY ON VALUE FUNCTION ESTIMATION

We analyze how different test-time adaptation strategies for incorporating temporal context impact value function estimation. VITA performs step-by-step adaptation without reset (implicit memory), incrementally updating parameters from θ_{t-1} . This allows the value function estimator to encode trajectory history directly into its parameters, capturing both temporal order and accumulated temporal context. We compare VITA with three alternatives: (i) *trajectory-level adaptation (TTT-TR)*, which applies a single gradient update computed over the full trajectory. This produces a batched update that averages the loss across all frames, thereby discarding temporal order; (ii) *memoryless adaptation (TTT-RS)*, which resets the adaptation module to θ_0 at every step. By adapting only to the current visual observation, this approach prevents any temporal information from being carried forward; (iii) *explicit memory (TTT-EX)*, which also resets the adaptation module at every step but updates using a local window of recent frames matching the window size used during meta-training ($w_{tr} = 8$). The margins here are the largest in the paper: across all ten datasets VITA outperforms trajectory-level, memoryless, and explicit-memory adaptation by roughly 0.5–0.6 VOC (e.g., 0.782 vs. at most 0.196 in-distribution; see Appendix F.2), while the three batched-update variants remain nearly indistinguishable from one another. This near-collapse of every non-sequential variant to a common low baseline isolates incremental, per-timestep accumulation of temporal context as the single ingredient responsible for VITA’s value estimation quality.

5 RELATED WORK

5.1 VLMS AS VALUE FUNCTION ESTIMATORS

Contrastive VLMS have been applied as zero-shot reward models (Baumli et al., 2023; Rocamonde et al., 2024) and goal-conditioned value functions (Ma et al., 2023b; 2024), using frame-level similarity scores but lacking temporal modeling; our experiments make the cost of this omission concrete, with such similarity-only baselines scoring near or below zero VOC under distribution shift. Some works fine-tune CLIP on domain-specific video datasets (Fan et al., 2022; Jiang et al., 2024) to generate dense rewards, requiring supervision that limits their zero-shot applicability. RoboCLIP (Sontakke et al., 2023) enables one-shot reward specification without fine-tuning, but yields sparse rewards and still requires demonstrations. Large-scale multimodal pre-training approaches (Alakuijala et al., 2024; Li et al., 2024; Ma et al., 2023b) learn language-conditioned value functions by relying on scale for generalization. ReWiND (Zhang et al., 2025b) trains a cross-modal sequential transformer on augmented large-scale data with video rewinding, combined with a few in-domain demonstrations to improve generalization. In contrast, VITA adapts a pre-trained contrastive vision–language encoder at inference, capturing both temporal and semantic context through test-time adaptation, without requiring large-scale multimodal pre-training or additional domain-specific demonstrations, and it does so while remaining the top performer across our generalization, discrimination, and reward-shaping evaluations. Because VITA is agnostic to the underlying encoder, it could enhance the generalization and temporal reasoning of any pre-trained vision–language representation that encodes a universal value function via meta-learned test-time adaptation. Autoregressive VLMS (Alayrac et al., 2022) leverage in-context learning over trajectories and have been applied to success detection (Du et al., 2023) and goal-conditioned value functions (Ma et al., 2024). GVL (Ma et al., 2024) mitigates monotonic bias from ordered pre-training data by shuffling frames at inference, but this discards temporal order, which is essential for temporal reasoning and for distinguishing visually similar yet temporally distinct states, and which we find leaves GVL trailing VITA on all ten datasets. Our approach instead preserves chronological order while capturing temporal context through test-time adaptation, avoiding such shortcuts.

5.2 TEST-TIME ADAPTATION

Test-time adaptation methods provide parameter-efficient solutions for adapting VLMS without full fine-tuning. Prompt-based methods optimize model parameters at test time based on the task context, although their underlying mechanisms are opaque (Zhou et al., 2022; Khattak et al., 2023; Ma et al., 2023a). Lim et al. (2025) suggested that CLIP representations could support test-time adaptation, based on the observation that CLIP encodes human-interpretable concepts discovered via mechanistic interpretability (Bricken et al., 2023). CLIP-based similarity has also been used as a

reward signal, with parameters optimized at test time via reinforcement learning (Zhao et al., 2024). Another approach is test-time training (Sun et al., 2020), where an adaptation module is updated at each timestep via a self-supervised loss. Unlike meta-reinforcement learning methods (Finn et al., 2017; Bauer et al., 2023), which require task-specific adaptation episodes during training, test-time training enables direct online adaptation during inference, without task labels, and, as our per-timestep ablation quantifies, doing so sequentially rather than in batched updates is what turns this paradigm into an effective temporal memory for value estimation.

6 DISCUSSION, LIMITATIONS, AND FUTURE WORK

6.1 DISCUSSION

Taken together, our three evaluations show that test-time adaptation lets value functions for robotic manipulation generalize across distribution shifts in task, environment, and embodiment, with VITA the best or tied-best method on each. It attains the top VOC on 6 of 10 shift datasets while never falling below the state-of-the-art autoregressive baselines, a perfect 1.00 BinVOC on expert-vs.-non-expert discrimination, and the highest offline-RL IQM return (0.815), together enabling downstream use in RL and imitation learning. By updating sequentially over a trajectory, VITA encodes history into its parameters and thereby captures temporal and semantic context more effectively than both CLIP-based baselines and in-context learning with autoregressive VLMs. The head-to-head margins over CLIP-GRU on discrimination (1.00 vs. 0.80) and offline RL (0.815 vs. 0.777) indicate that implicit memory via sequential test-time updates generalizes better than encoding temporal history in recurrent hidden states. Pre-trained autoregressive VLMs (GVL), despite broad generalization, are not explicitly trained for progress estimation, which their consistently lower VOC reflects. Finally, these gains come at negligible inference-time cost, since only a lightweight adaptation module is updated per timestep, so real-time applicability is preserved.

6.2 LIMITATIONS AND FUTURE WORK

Although test-time adaptation improves zero-shot progress estimation across unseen tasks and environments in our experiments, the very setting where the margins are widest, long-horizon execution, is also where absolute VOC is lowest for every method (e.g., `ms_sweep` at 0.490 for VITA), indicating that high execution variability and extended durations remain challenging in absolute terms even as VITA leads. In addition, while the adaptation overhead is negligible thanks to the lightweight module, updating a value function estimator at every timestep may be unsafe during deployment, which could limit applicability in some real-time scenarios. In future work, we plan to explore alternative test-time adaptation schemes for vision-language models, particularly for training agents within world models, and to apply VITA to real-time closed-loop control and more complex RL environments. Because VITA is agnostic to the underlying vision-language encoder, it could enhance generalization and temporal reasoning for any domain-specific pre-trained multimodal representation via meta-learned test-time adaptation. Finally, while we provide empirical evidence that dissimilarity-based sampling mitigates shortcut learning, most directly the jump from 0.20 to 1.0 Disc@VOC in our ablation, a theoretical account of how diversity-based sampling influences shortcut learning remains open. This lies beyond the scope of our core contribution: a value function estimator that meta-learns a test-time adaptation mechanism to improve zero-shot performance and temporal reasoning.

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A TRAINING DATASET

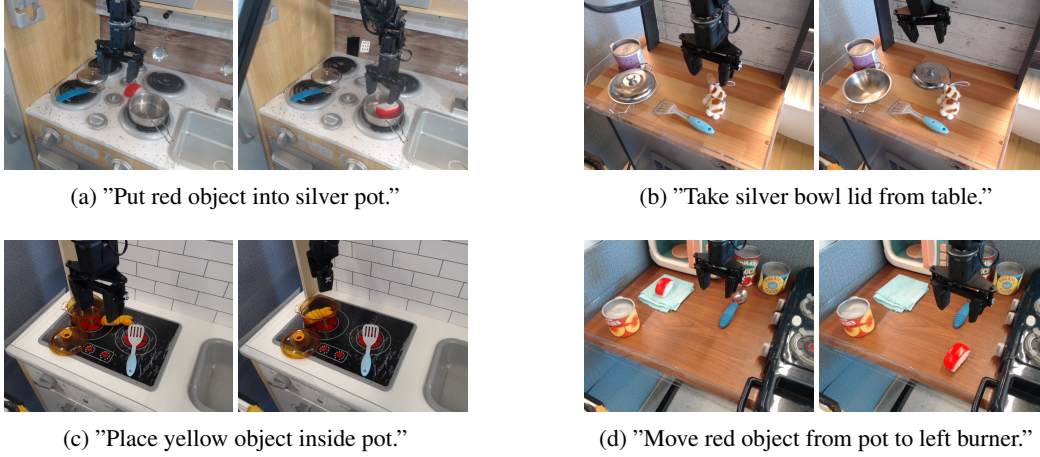


Figure 4: Start and end frames of representative expert demonstrations used for training, each with its natural language task description. All examples are pick-and-place tasks collected across the four ToyKitchen configurations that make up the entire training distribution, underscoring how narrow the training set is relative to the ten out-of-distribution datasets used for evaluation.

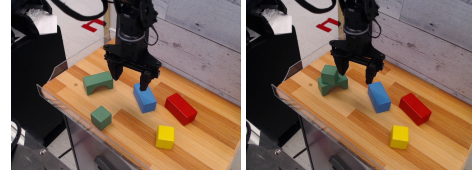
B EVALUATION DATASETS

Table 3: The ten evaluation datasets, listed by task type, environment, and embodiment. Only `tk_pnp` matches the training distribution; the remaining nine each introduce at least one shift (new environment, new embodiment, or a task type such as folding, sweeping, or stacking never seen in training), which is what the VOC comparison in Table 1 is measured against.

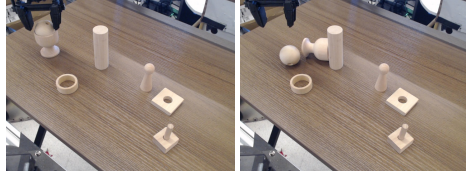
Dataset	Task Type	Environment	Embodiment
<code>tk_pnp</code>	pick-and-place	toy kitchen	WidowX 250
<code>lm_pnp</code>	pick-and-place	laundry machine	WidowX 250
<code>td_fold</code>	fold cloth	tabletop (dark wood)	WidowX 250
<code>ft_fold</code>	fold cloth	folding table	WidowX 250
<code>rd_fold</code>	fold cloth	robot desk	WidowX 250
<code>ms_sweep</code>	sweep	folding table (tray)	WidowX 250
<code>dt_tk_pnp</code>	pick-and-place	toy kitchen	DeepThought
<code>dt_tk_stack</code>	stack blocks	toy kitchen	DeepThought
<code>dt_ft_stack</code>	stack blocks	folding table	DeepThought
<code>dt_rd_pnp</code>	pick-and-place	robot desk (drawer)	DeepThought



(a) dt_tk_ms: "Move yellow object from burner to center."



(b) dt_tk_stack: "Place green cube on top of arch."



(c) dt_ft_stack: "Move egg to table."



(d) dt_rd_pnp: "Put orange object in drawer."

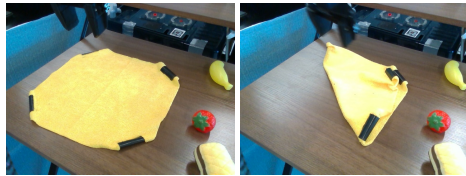
Figure 5: Start and end frames of evaluation trajectories under embodiment shift, each with its natural language task description. The top row keeps the training environment (ToyKitchen) but swaps in a different robot (DeepThought); the bottom row compounds the embodiment change with a new environment. These are exactly the EM and ES & EM rows on which VITA sustains VOC of 0.695–0.820 in Table 1.



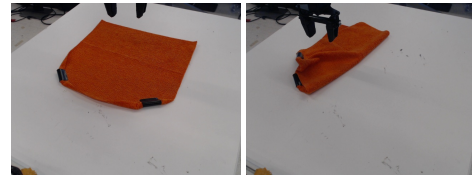
(a) lm_pnp: "Place blue cloth inside washer."



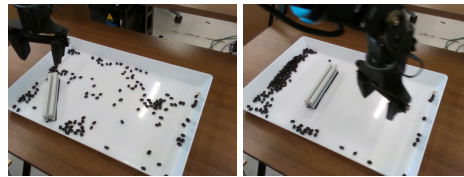
(b) td_fold: "Fold cloth bottom right to top left."



(c) ft_fcnp: "Fold clothes from left to right."



(d) rd_fold: "Fold cloth bottom right to top left."



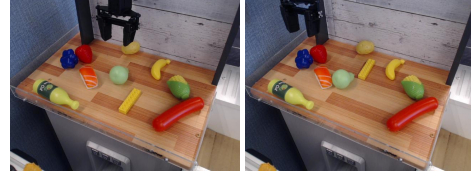
(e) ms_ft_sweep: "Sweep into pile."

Figure 6: Start and end frames of evaluation trajectories under environment shift, each with its natural language task description. The robot embodiment is held fixed (WidowX 250) while the environment varies substantially in appearance and structure, isolating environment shift as the sole source of variation, the ES rows where VITA still attains VOC of 0.490–0.725 in Table 1.

C SCRIPTED DATASET



(a) `sc_pnp_obj`: "Pick up an object and place it."



(b) `sc_pnp_robj`: "Pick up a rigid object and place it."



(c) `sc_pnp_stoy`: "Pick up a soft object and place it."



(d) `sc_pnp_uten`: "Pick up a kitchen utensil and place it."



(e) `sc_pnp_cloth`: "Move the cloth."

Figure 7: Start and end frames of scripted (non-expert) evaluation trajectories collected in the ToyKitchen environment with the WidowX 250 robot. All are pick-and-place behaviors across varying object categories, matching the training distribution in task, embodiment, and environment so that the only meaningful difference from expert data is behavioral quality, exactly the axis the BinVOC metric in Table 2a scores. Generic task descriptions are used due to the lack of publicly available annotations.

D MODEL ARCHITECTURE

D.1 MODEL AND TRAINING DETAILS

We use the OpenCLIP ViT-B/32 encoder pre-trained on the OpenAI dataset as a frozen backbone, encoding each visual observation and task description with the CLIP vision and text encoders respectively. The joint (concatenation-based) representation was chosen over an element-wise product because it consistently yielded higher validation VOC. The adaptation module f_{adapt} is a two-layer residual MLP with GELU activation and projection dimension $d' = 64$. Training uses AdamW with learning rate 1×10^{-4} , weight decay 1×10^{-4} , and a cosine schedule with 10% warmup, batch size 32, and all trajectories padded to a maximum length of 120 frames (the longest sequence in the training set). The self-supervised weighting coefficient λ_{self} is set to 0.5, again selected by validation VOC. We train for only 5 epochs, since validation VOC plateaus early and longer training yields no measurable improvement. For dissimilarity-based sampling we use stride $s = 1$, sub-trajectory length $w_{\text{tr}} = 8$, and $k = 8$ selected windows unless otherwise specified, the configuration that reaches perfect discrimination in Table 5. All experiments ran on NVIDIA RTX 6000 Ada Generation GPUs.

D.2 TEST-TIME TRAINING HYPERPARAMETERS

At inference time we adapt only the temporal adaptation module f_{adapt} , using the same projection dimension ($d' = 64$) as in training, with a single gradient step ($t_{\text{ep}} = 1$) at learning rate $\eta = 0.1$.

Table 4: Validation VOC as a function of the test-time learning rate η and number of adaptation steps t_{ep} . Best VOC in each column is in **bold**. Two well-separated optima emerge, a single large-step update ($\eta = 0.1$, $t_{\text{ep}} = 1$; 0.783) and many small-step updates ($\eta = 0.01$, $t_{\text{ep}} = 10$; 0.781), which are essentially tied; we adopt the former for its lower cost.

η	$t_{\text{ep}} = 1$	$t_{\text{ep}} = 5$	$t_{\text{ep}} = 10$
0.001	0.225	0.345	0.453
0.01	0.454	0.744	0.781
0.05	0.746	0.670	0.515
0.1	0.783	0.512	0.369
1.0	0.299	0.183	0.181
5.0	0.169	0.182	0.157

This choice follows a sweep over learning rates $\{5.0, 1.0, 0.1, 0.05, 0.01, 0.001\}$ and adaptation steps $\{1, 5, 10\}$ (Table 4). The sweep reveals two comparably strong optima: a single large-step update ($\eta = 0.1$, $t_{\text{ep}} = 1$; VOC 0.783) and ten small-step updates ($\eta = 0.01$, $t_{\text{ep}} = 10$; VOC 0.781), a difference of only 0.002. We adopt the single-step setting because it matches the best VOC while requiring one-tenth the adaptation steps. Performance is also stable near this optimum but degrades steeply at extreme learning rates (VOC falls to 0.169 at $\eta = 5.0$), confirming that the selected setting sits in a well-behaved region.

D.3 BASELINES

To keep the comparison fair, all learned baselines share as much of VITA’s design as possible and differ only along the axis under study. VITA uses $t_{\text{ep}} = 1$, projection dimension $d' = 64$, and learning rate $\eta = 0.1$, selected via the tuning described above. CLIP-FT follows VITA’s architecture but removes the adaptation module and self-supervised loss, isolating supervision-without-temporal-modeling: a frozen CLIP encoder feeds a linear projection and a two-layer MLP trained with standard supervised regression, no meta-learning or test-time adaptation. To make this an especially strong baseline rather than a handicapped one, CLIP-FT is given an $8\times$ larger projection matrix and $10\times$ more training steps than VITA (Lin et al., 2022); that it still fails to generalize (Table 1) therefore cannot be attributed to insufficient capacity or training. For VLM-RM we use a baseline prompt describing the environment. CLIP-GRU again follows VITA’s architecture (frozen CLIP encoder and regression head h) but replaces f_{adapt} with a single-layer GRU of hidden size 64 whose final hidden state feeds h , isolating recurrent hidden states as the temporal-memory mechanism, the cleanest contrast with VITA’s implicit parameter memory. For both GVL-0S and GVL-1S we use the latest Gemini 1.5 Pro, `gemini-1.5-pro-latest` (Gemini Team et al., 2024), a newer release than the `gemini-1.5-pro` used in the original GVL. We additionally evaluated the open-source Qwen-VL 2.5 (Yang et al., 2024), which failed to overcome temporal bias despite frame shuffling, while GPT-4o (Hurst et al., 2024) consistently declined to produce scalar progress estimates. All GVL evaluations used the prompt template from the original work (Ma et al., 2024).

E OFFLINE RL SETUP

E.1 TRAINING CONFIGURATION

To ensure that differences in MT10 return reflect reward quality rather than RL tuning, we use Implicit Q-Learning (IQL) with a single fixed configuration across all tasks and all reward sources. Expectile regression is set to $\tau = 0.7$, with advantage weighting temperature 3.0 and clipping threshold 100.0. The actor and critic use learning rates 1×10^{-4} and 3×10^{-4} , respectively, with batch size 256. Policies are trained for 100,000 gradient steps, with evaluation every 10,000 steps on 20 rollouts per task (horizon 150). Implementation follows `d3rlpy`, with configuration: `expectile=0.7`, `weight_temp=3.0`, `max_weight=100.0`, `actor_lr=1e-4`, `critic_lr=3e-4`, `batch_size=256`.

E.2 EVALUATION PROTOCOL

Following Agarwal et al. (Agarwal et al., 2021), we report the interquartile mean (IQM) of returns across all task–seed pairs. The IQM averages the middle 50% of scores and is therefore less sensitive to outliers than the mean or median, giving a more reliable point of comparison across methods. We accompany each IQM with a 95% confidence interval estimated by stratified bootstrap with 10,000 resamples, stratified by task, so that the margins reported in Table 2b, including VITA’s non-overlapping lead over META-WL, can be read directly from the intervals. The identical procedure is applied to every method in the Meta-World MT10 experiments.

F ABLATION STUDIES

F.1 EFFECT OF DISSIMILARITY-BASED SAMPLING ON DISCRIMINATIVE PERFORMANCE

Table 5: Per-dataset Disc@VOC under different training-time sampling strategies (a checkmark marks successful expert-vs.-suboptimal discrimination), with the average over the five scripted datasets in the bottom row. Sampling methods **FT** (full-trajectory), **Rand** (random), and **Diss** (dissimilarity-based) are parameterized by (w_{tr}, k) . Dissimilarity-based sampling at (8, 8) is the only setting to discriminate on all five datasets (avg. 1.0), a $5\times$ improvement over full-trajectory sampling (0.20) and a clear gain over random sampling (0.80).

Dataset	FT (8,–)	Rand (8,8)	Diss (8,4)	Diss (4,8)	Diss (8,8)
sc.pnp_obj		✓	✓		✓
sc.pnp_robj	✓	✓	✓	✓	✓
sc.pnp_stoy			✓		✓
sc.pnp_uten	✓	✓	✓	✓	✓
sc.pnp_cloth		✓	✓	✓	✓
Avg. Disc@VOC	0.20	0.8	1.0	0.60	1.0

We validate dissimilarity-based sampling by holding the model fixed and varying only the training-time sub-trajectory sampling strategy, then measuring how well the resulting model discriminates expert from suboptimal trajectories. Table 5 reports Disc@VOC over 5 scripted datasets, i.e., whether the model assigns higher average VOC to expert trajectories (τ_{k_pnp}) than to each scripted dataset. We compare three strategies: (1) full-trajectory sampling, which adapts over all overlapping sub-trajectories of length w_{tr} with stride s , following prior work (Sun et al., 2024); (2) random sampling, which uniformly draws k sub-trajectories of length w_{tr} per video; and (3) our dissimilarity-based sampling, which builds a candidate set of length- w_{tr} sub-trajectories with stride s and selects the k that maximize pairwise dissimilarity in representation space.

The comparison yields a clear ranking. Dissimilarity-based sampling with $w_{tr} = 8$ and $k = 8$ is the only configuration to discriminate on all five datasets (Disc@VOC 1.0); reducing to $k = 4$ at the same window length holds this perfect score, whereas shrinking the window to $w_{tr} = 4$ drops it to 0.60. Full-trajectory sampling is worst at 0.20, confirming that adapting over all consecutive windows overfits to global temporal cues, and random sampling lands in between at 0.80, better than full-trajectory but still one dataset short of dissimilarity-based selection. Together these numbers show that both sub-trajectory diversity and an adequate window length are needed for discriminative test-time adaptation.

F.2 EFFECT OF TEMPORAL MEMORY ON VALUE FUNCTION ESTIMATION

We evaluate four adaptation strategies: (1) TTT-TR, which performs a single gradient update computed over the full trajectory (offline). TTT-TR performs a batched update that averages the loss across all frames, thereby discarding temporal order; (2) TTT-RS, which resets the adaptation module at every step and updates using only the current visual observation (memoryless). TTT-RS prevents any temporal information from being carried forward; (3) TTT-EX, which resets the adaptation module at every step and updates using a local window of recent frames matching the window size used during meta-training ($w_{tr} = 8$). TTT-EX averages gradients over this window, discarding the

Table 6: Validation VOC for progress estimation under distribution shifts, comparing three batched-update TTT variants against VITA’s sequential per-timestep updates; best per row in **bold**. VITA wins every row by a wide margin (roughly 0.5–0.6 VOC), while TTT-TR, TTT-RS, and TTT-EX stay clustered near 0.1–0.2 and turn negative on `ms_sweep`. Shift regimes: ID = In-Distribution, ES = Environment Shift, EM = Embodiment Shift, ES+EM = Environment and Embodiment Shift.

Shift	Dataset	TTT-TR	TTT-RS	TTT-EX	VITA
ID	tk_pnp	0.1917	0.1918	0.1957	0.7822
ES	lm_pnp	0.1843	0.1854	0.1826	0.7246
	td_fold	0.1402	0.1393	0.1322	0.7085
	ft_fold	0.1302	0.1311	0.1683	0.6583
	rd_fold	0.1161	0.1172	0.1154	0.6056
	ms_sweep	-0.0225	-0.0191	0.0102	0.4898
EM	dt_tk_pnp	0.2123	0.2118	0.2069	0.8203
	dt_tk_stack	0.0833	0.0830	0.0747	0.7081
ES+EM	dt_ft_stack	0.0558	0.0537	0.0439	0.6979
	dt_rd_pnp	0.1665	0.1660	0.1559	0.6951

temporal ordering within it; (4) VITA (OURS), which updates the adaptation module incrementally at every timestep (implicit memory), forming an implicit memory through the sequential accumulation of temporal history in the adapter parameters. To keep the comparison fair, all four variants received the same test-time hyperparameter sweep (Appendix D); VITA, TTT-RS, and TTT-TR were best at learning rate 0.1 with a single step, and TTT-EX at learning rate 1.0 with one step.

Table 6 draws a stark line between the two families. The three batched-update variants, TTT-TR, TTT-RS, and TTT-EX, are effectively interchangeable, clustering within roughly 0.02 of one another on every dataset (e.g., 0.19–0.20 in-distribution) and even going slightly negative on `ms_sweep`, so batching the update, whether over a short explicit window (TTT-EX) or the full trajectory (TTT-TR), buys nothing over the memoryless baseline (TTT-RS). VITA instead scores 0.490–0.820 across the same datasets, beating each batched variant by about 0.5–0.6 VOC on every row, the largest and most uniform margins reported anywhere in the paper. Two mechanisms explain this gap. Updating over a local window would double-count the most recent frames, once explicitly through the window and once implicitly through the accumulated parameters, while updating less frequently would discard intermediate frames entirely. The implicit-memory paradigm therefore depends on sequential, per-timestep updates without reset, and the numbers show that this dependence is decisive rather than incidental.

G COMPLEXITY ANALYSIS OF DISSIMILARITY-BASED SAMPLING

Let $\mathcal{W}$ denote the candidate set of size $N_c = |\mathcal{W}|$. To approximate diverse subset selection without the exponential cost of enumerating all $\binom{N_c}{b}$ subsets as described in Eq. 4, we adopt a scoring-based heuristic that assigns each candidate window a diversity score equal to the sum of its pairwise distances to all others (i.e., the row sum of the distance matrix). Let the number of candidate windows be $N_c = T - w_{tr} + 1$, each represented by a flattened feature vector of dimension $m = w_{tr} \cdot d$, where d is the dimension of each multimodal representation z_t . We compute the full pairwise distance matrix $D \in \mathbb{R}^{N_c \times N_c}$ on the GPU using batched `cdist`, which requires $\mathcal{O}(N_c^2 \cdot w_{tr} \cdot d)$ operations.

In our experiments, training trajectories are short (mean length 69, max length 120). With a batch size of 32, feature dimension $d = 1024$ (CLIP-based multimodal representation), and window length $w_{tr} = 8$, the worst-case computational cost is:

$$32 \cdot N_c^2 \cdot w_{tr} \cdot d = 32 \cdot 114^2 \cdot 8 \cdot 1024 \approx 340 \text{ MFLOPs.}$$

At roughly 340 MFLOPs in the worst case, this overhead is negligible next to a single forward-and-backward pass through the model, so dissimilarity-based sampling delivers its perfect-discrimination gains (Table 5) at essentially no added training cost.